Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Assignment #4 - Heat Equation
Due Tuesday, October 20, 2026

1. (10 points) Let $k, L > 0$ and $f \in \mathrm{PC}^1[0, L]$ be a given function. Consider the heat equation

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{k}\frac{\partial u}{\partial t} = 0. \tag{HE}$$

(a) Write down the solution to (HE) with the boundary conditions

$$\frac{\partial u}{\partial x}(t, 0) = \frac{\partial u}{\partial x}(t, L) = 0, \qquad t > 0$$

and the initial condition

$$u(0, x) = f(x), \qquad 0 < x < L.$$

Solution — As with the wave equation, we apply an identical strategy of assuming the solution takes the separated form

$$u(t, x) = T(t)X(x)$$

Then

$$\frac{\partial^2 u}{\partial x^2} = T(t)X''(x) \qquad \frac{\partial u}{\partial t} = T'(t)X(x)$$

and plugging into the heat equation, we obtain:

$$k\frac{X''(x)}{X(x)} = \frac{T'(t)}{T(t)} = -\lambda$$

The left-hand side is a function of x only and the right-hand side is a function of t only. For them to be equal to one another for all values of t, x, they must both be equal to the same constant, which (for future convenience) we will notate as $-\lambda$. Hence our separated heat equation reduces to a system of independent, linear, homogeneous ordinary differential equations:

$$\begin{cases} kX''(x) + \lambda X(x) = 0 \\ T'(t) + \lambda T = 0 \end{cases}$$

We see that the structure of solutions to these equations depends on the sign of λ, so

we split into cases accordingly:

For $\lambda < 0$:

$$T(t) = c_0 e^{-\lambda t} \qquad X(x) = c_1 e^{\sqrt{-\frac{\lambda}{k}}x} + c_2 e^{-\sqrt{-\frac{\lambda}{k}}x}$$

For $\lambda = 0$:

$$T(t) = c_3 \qquad X(x) = c_4 + c_5 x$$

For $\lambda > 0$:

$$T(t) = c_6 e^{-\lambda t} \qquad X(x) = c_7 \cos\left(\sqrt{\frac{\lambda}{k}}x\right) + c_8 \sin\left(\sqrt{\frac{\lambda}{k}}x\right)$$

We now constrain these general solutions with the homogenous Neumann boundary conditions:

$$\frac{\partial u}{\partial x}(t,0) = 0 \implies T(t)X'(0) = 0 \implies X'(0) = 0$$
$$\frac{\partial u}{\partial x}(t,L) = 0 \implies T(t)X'(L) = 0 \implies X'(L) = 0$$

For $\lambda < 0$:

$$X'(0) = 0 \implies c_1\sqrt{-\frac{\lambda}{k}} - c_2\sqrt{-\frac{\lambda}{k}} = 0 \implies \boxed{c_1 = c_2}$$
$$X'(L) = 0 \implies c_1\sqrt{-\frac{\lambda}{k}}e^{\sqrt{-\frac{\lambda}{k}}L} - c_2\sqrt{-\frac{\lambda}{k}}e^{-\sqrt{-\frac{\lambda}{k}}L} = 0$$
$$\implies c_1\sqrt{-\frac{\lambda}{k}}\left(e^{\sqrt{-\frac{\lambda}{k}}L} - e^{-\sqrt{-\frac{\lambda}{k}}L}\right) = 0 \quad (\text{since } c_1 = c_2)$$
$$\implies \boxed{c_1 = 0} \quad (\text{Exponential factor} \neq 0 \text{ for } \lambda \neq 0)$$

For $\lambda = 0$:

$$X'(0) = X'(L) = 0 \implies \boxed{c_5 = 0}$$

For $\lambda > 0$:

$$
\begin{aligned}
X'(0) = 0 &\implies c_8\sqrt{\frac{\lambda}{k}} = 0 \\
&\implies \boxed{c_8 = 0} \quad (\text{since } \lambda > 0, k > 0) \\
X'(L) = 0 &\implies -c_7\sqrt{\frac{\lambda}{k}}\sin\left(\sqrt{\frac{\lambda}{k}}L\right) = 0 \\
&\implies \sin\left(\sqrt{\frac{\lambda}{k}}L\right) = 0 \quad (\text{since } \lambda \neq 0, k \neq 0, \text{ and assuming } c_7 \neq 0)
\end{aligned}
$$

Note that $c_7 = 0$ would produce a trivial solution since c_8 already equals zero, and the trivial solution is already accounted for in the other λ-cases.

$$
\begin{aligned}
&\implies \sqrt{\frac{\lambda}{k}}L = n\pi \quad (\text{for any } n \in \mathbb{Z}) \\
&\implies \boxed{\lambda = k\left(\frac{n\pi}{L}\right)^2}
\end{aligned}
$$

We write the general solutions produced by each case (with the Neumann boundary conditions applied):

For $\lambda < 0$:

$$
c_1 = c_2 = 0 \implies X(x) = 0 \implies u(t,x) = 0
$$

For $\lambda = 0$:

$$
c_5 = 0 \implies X(x) = c_4 \implies u(t,x) = c_3c_4 = C_0 \quad \text{for some } C_0 \in \mathbb{R}
$$

For $\lambda > 0$:

$$
c_8 = 0 \implies u_n(t,x) = C_n e^{-k\left(\frac{n\pi}{L}\right)^2 t}\cos\left(\frac{n\pi x}{L}\right) \quad \text{for } C_n \in \mathbb{R}
$$

Note that each separated eigenfunction solution will have its own constant C_n. Hence the general solution is the superposition of all solutions from each of these 3 cases:

$$
\boxed{u(t,x) = C_0 + \sum_{n=1}^{\infty} C_n e^{-k\left(\frac{n\pi}{L}\right)^2 t}\cos\left(\frac{n\pi x}{L}\right)}
$$

Let us now apply the initial condition $u(0,x) = f(x)$ for $0 \leq x \leq L$.

$$\implies C_0 + \sum_{n=1}^{\infty} C_n \cos\left(\frac{n\pi x}{L}\right) = f(x)$$

Notice that the application of this initial condition creates an equation that is exactly the Fourier series expression of $f(x)$. For both sides to be equal, we must have that each C_n is exactly the nth Fourier coefficient in the expansion of $f(x)$ on $[0, L]$. Hence

$$\boxed{C_0 = \frac{1}{L}\int_0^L f(x)dx}$$

$$\boxed{C_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx \qquad n \geq 1}$$

(b) Let $T_1 > 0$, take

$$f(x) = T_1\frac{x}{L},$$

and determine the resulting solution by computing the Fourier coefficients and the corresponding Fourier series.

Solution —

$$\begin{aligned} C_0 &= \frac{1}{L}\int_0^L \left(T_1\frac{x}{L}\right)dx \\ &= \frac{T_1}{L^2}\int_0^L x dx \\ &= \frac{T_1}{L^2}\left(\frac{1}{2}L^2\right) \\ &= \frac{T_1}{2} \end{aligned}$$

For the general coefficients

$$\begin{aligned} C_n &= \frac{2}{L}\int_0^L \left(T_1\frac{x}{L}\right)\cos\left(\frac{n\pi x}{L}\right)dx \\ &= \frac{2T_1}{L^2}\int_0^L x\cos\left(\frac{n\pi x}{L}\right)dx \end{aligned}$$

Apply integration by parts:

$$\begin{aligned} U &= x & dV &= \cos\left(\frac{n\pi x}{L}\right)dx \\ dU &= dx & V &= \frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right) \end{aligned}$$

$$
\begin{aligned}
C_n &= \frac{2T_1}{L^2}\left[\underbrace{\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{x=0}^{x=L}}_{0} - \frac{L}{n\pi}\int_0^L \sin\left(\frac{n\pi x}{L}\right)dx\right] \\
&= \frac{-2T_1}{n\pi L}\left[-\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_{x=0}^{x=L} \\
&= \frac{2T_1}{(n\pi)^2}\left[\cos\left(\frac{n\pi x}{L}\right)\right]_{x=0}^{x=L} \\
&= \frac{2T_1}{(n\pi)^2}\left[\cos(n\pi) - 1\right] \\
&= \frac{2T_1}{(n\pi)^2}\left[(-1)^n - 1\right]
\end{aligned}
$$

Hence the solution is

$$
\boxed{u(t,x) = \frac{T_1}{2} + \frac{2T_1}{\pi^2}\sum_{n=1}^{\infty}\left[\frac{(-1)^n - 1}{n^2}\right]e^{-k\left(\frac{n\pi}{L}\right)^2 t}\cos\left(\frac{n\pi x}{L}\right)}
$$

2. (15 points) Let $L, T_0 > 0$ and $f \in \mathrm{PC}^1[0, L]$ be given. Consider the temperature distribution of a long, thin bar in which heat is internally generated by a source with constant rate. In particular, we let $h > 0$ be given and consider the heat equation

$$
\frac{1}{k}\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + h
$$

for $0 < x < L$ and $t > 0$ with given boundary conditions

$$
u(t, 0) = T_0 \qquad \text{and} \qquad \frac{\partial u}{\partial x}(t, L) = 0
$$

and initial temperature distribution

$$
u(0, x) = f(x).
$$

Determine the temperature throughout the bar at all later times, namely compute $u(t, x)$.

Solution — Because this is the inhomogeneous heat equation (caused by the presence of the h term), we cannot separate the equation that results from assuming $u(t, x) = T(t)X(x)$ and plugging in. Hence we need a different method—namely, the one that involves finding a steady-state solution.

Assume the bar settles on the steady-state distribution $u_s(x)$ as $t \to \infty$. Because this

particular variation of the heat equation should be valid for all time, it's definitely valid for $u(t,x) = u_s(x)$ at $t = \infty$. Hence, plugging in $u_s(x)$, we obtain

$$\begin{aligned} 0 &= u_s''(x) + h \\ \implies u_s''(x) &= -h \\ \implies u_s'(x) &= -hx + C_0 \\ \implies u_s(x) &= -\frac{1}{2}hx^2 + C_0x + C_1 \end{aligned}$$

We can apply the boundary conditions to the steady-state solution:

$$\begin{aligned} u(t,0) = T_0 &\implies u_s(0) = T_0 \\ &\implies C_1 = T_0 \\ \frac{\partial u}{\partial x}(t,L) = 0 &\implies u_s'(L) = 0 \\ &\implies -hL + C_0 = 0 \\ &\implies C_0 = hL \end{aligned}$$

Hence the steady-state solution with applied boundary conditions is

$$u_s(x) = -\frac{1}{2}hx^2 + hLx + T_0$$

We can now perform a change-of-variable transformation that uses this steady-state solution to transform the given heat equation into a homogeneous (and separable) form. Namely, let $v(t,x)$ be the difference in heat between the current heat $u(t,x)$ and the eventual final heat $u_s(x)$:

$$v(t,x) = u(t,x) - u_s(x)$$

We plug this new expression of $u(t,x)$ into the given heat equation (Note that $u = v + u_s$)

$$\frac{1}{k}\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2} + u_s''(x) + h$$

And note that $u_s''(x) = -h$, so we recover the homogeneous form

$$\frac{1}{k}\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$$

With boundary conditions

$$\begin{aligned} u(t,0) = T_0 &\implies v(t,0) + \underbrace{u_s(0)}_{T_0} = T_0 \\ &\implies v(t,0) = 0 \end{aligned}$$

$$\frac{\partial u}{\partial x}(t,L)=0 \implies \frac{\partial v}{\partial x}(t,L)+\underbrace{u_s'(L)}_{0}=0$$
$$\implies \frac{\partial v}{\partial x}(t,L)=0$$

And initial condition

$$u(0,x)=f(x) \implies v(0,x)+u_s(x)=f(x)$$
$$\implies v(0,x)=f(x)-u_s(x)$$

Since we've transformed the problem to a homogeneous form in $v(t,x)$, we proceed to solve regularly in v, assuming $v(t,x)=T(t)X(x)$. Then

$$k\frac{X''(x)}{X(x)}=\frac{T'(t)}{T(t)}=-\lambda$$

$$\begin{cases} kX''(x)+\lambda X(x)=0 \\ T'(t)+\lambda T=0 \end{cases}$$

3. (13 points) Let c be a given number satisfying $\frac{\pi}{10}<c<\frac{9\pi}{10}$ and define for every $0<x<\pi$ the function

$$h(x)=\begin{cases} 5, & \text{if } c-\frac{\pi}{10}<x<c+\frac{\pi}{10} \\ 0, & \text{else.} \end{cases}$$

(a) Solve the heat equation with $k=1$, namely

$$\frac{\partial^2 u}{\partial x^2}-\frac{\partial u}{\partial t}=0$$

with

$$u(t,0)=u(t,\pi)=0, \qquad t>0$$

and

$$u(0,x)=h(x), \qquad 0<x<\pi.$$

Use the trigonometric identity

$$\cos(\alpha)-\cos(\beta)=-2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

to simplify the form of the Fourier coefficients.

(b) Notice that for t sufficiently large, we have

$$u(t,x)\approx B_1 e^{-t}\sin(x).$$

What choice of c as defined above makes B_1 largest?

(c) Using the idea in the previous part, answer the following question: Suppose that you're forced to put a cup of hot coffee somewhere between two blocks of ice. Where exactly should you put it if you want it to stay as warm as possible?

4. (12 points) Let $f, g \in \mathrm{PC}^1[0, \pi]$ be given and find the solution of the inhomogeneous wave equation with speed of propagation $c = 1$, namely

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + 1 = 0$$

for $0 < x < \pi$ and $t > 0$ subject to the given boundary conditions

$$u(t, 0) = 0 \qquad \text{and} \qquad u(t, \pi) = 0$$

for $t > 0$ and initial conditions

$$u(0, x) = f(x) \qquad \text{and} \qquad \frac{\partial u}{\partial t}(0, x) = g(x)$$

for $0 < x < \pi$.